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SUPERSEDING
MIL–G–15342(SHIPS)
15 October 1950

MILITARY SPECIFICATION

GASKETS (METALLIC ASBESTOS, SPIRAL WOUND, SYMBOL 2410, FOR BOILERS)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 Scope. — This specification covers metallic-asbestos gaskets for handholes and manholes of boilers on board Naval vessels.

1.2 Classification.—Gaskets shall be of the following classes as specified (see 6.1):

Class a—Boiler handhole.

Class b—Boiler manhole.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN–B–621 — Boxes, Wood, Nailed and Lock-Corner.
NN–B–631 — Boxes, Wood, Wirebound, for Domestic Shipment.
QQ–S–781—Strapping, Flat; Steel.

MILITARY

JAN–P–106 — Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.
JAN–B–107 — Boxes, Wood, Wirebound (Overseas Type).

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking of Shipments.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. — Metallic-asbestos gaskets furnished under this specification shall be a product which has been tested and has passed the qualification tests specified in section 4 (see 6.2).

3.2 Material. — The material shall be as specified herein.

3.3 Construction.—Gaskets shall be composed of a single strip of metal wound in spiral dovetail shape, starting on the inside

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and working outward ply by ply to the desired size. Between the plies of metal shall be a cushion of asbestos paper, in one or more sections, spirally wound to conform with the metal strip.

3.4 Metal strip.

3.4.1 *Composition.* — Unless otherwise specified in the contract or order, the metal strip (see 3.4.2) shall be of corrosion-resisting steel of the following composition:

Element	Percent
Chromium (min.)	18.0
Nickel (min.)	8.0
Carbon (max.)	0.1

3.4.2 *Thickness.* — The metal strip (see 3.4.1) shall be not less than 0.007 inch and not more than 0.015 inch thick.

3.5 Gasket thickness.—The finished gasket over the metal shall be 0.175 ± 0.005 inch thick.

3.6 Pressure. — Gaskets shall be suitable for pressures up to 600 p.s.i., as specified (see 6.1).

3.7 Endurance. — Gaskets shall pass the endurance test specified in 4.5.2 and 4.5.3.

3.8 Class a, boiler handhole.

3.8.1 *Dimensions and number of plies.*— Unless otherwise specified in the contract or order, class a gaskets shall conform to the dimensions and have the number of plies of metal shown in table I.

3.8.2 *Dimensional tolerances.* — Dimensional tolerances as shown in table II will be permitted.

TABLE I.—*Dimensions and number of plies.*

Description	Inside dimensions	Final width overall	Bearing Surface	Number of plies[1]	Boiler used with
	Inches	*Inch*	*Inch*		
Oval[2]	5-1/2 by 4-9/16	3/16	1/8	7	B&W, CE
Oval[3]	3-3/4 by 3-3/8	3/16	1/8	7	B&W, CE
Rectangular[4]	5 by 4-13/16	7/32	5/32	7	B&W, CE
No. 48 oval	4-23/32 by 3-25/32	7/32	5/32	7	B&W
4 inch tube oval	5-3/16 by 4-3/16	5/16	1/4	8	FW
2-3/4 by 3-1/4 inches oval[5]	3-13/32 by 2-25/32	7/32	5/32	7	FW, CE
3-5/16 inch plug oval	3-31/32 by 3-11/32	7/32	5/32	7	FW
No. 21 oval	2-1/2 by 2-1/8	3/16	1/8	7	CE
No. 24 oval	3-7/8 by 3	19/64	15/64	8	CE
No. 28 oval	3-7/8 by 3-3/8	7/32	5/32	7	CE
No. 29 oval	4-1/4 by 3-3/8	9/32	7/32	7	CE
No. 30 oval	4-1/2 by 3-5/8	9/32	7/32	7	CE
No. 33 oval	4-5/8 by 3-3/4	1/4	3/16	8	CE

[1]For gasket sizes other than these, the number of plies shall be at rate of 36 per inch of the bearing surface.
[2]Used with No. 24 B&W handhole plate and No. 32 CE handhole plate.
[3]Used with No. 40 and 44 B&W handhole plate and No. 27 CE handhole plate.
[4]Used with No. 28 B&W handhole plate and No. 51 CE handhole plate.
[5]Used with 2-3/4 by 3-1/4 inch FW plug and No. 23 CE handhole plate.

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TABLE II.—*Dimensional tolerances.*

Inside Dimension		Outside dimension	
Plus	Minus	Plus	Minus
Inch 1/64	*Inch* 0	*Inch* 0	*Inch* 1/64

3.9 Class b, boiler manhole.

3.9.1 *Dimensions.*—Class b gaskets shall be ½ inch wide overall and have a 7/16-inch bearing surface. The inside of the gasket shall follow the contour of the manhole cover and have a clearance of 1/32 inch between the cover and the inside of the gasket at all points.

3.9.2 *Number of plies.*—Gaskets shall contain not less than 42 plies of metal per inch of cross section of the bearing surface.

3.9.3 *Dimensional tolerances.* — Dimensional tolerances as shown in table III will be permitted.

TABLE III.—*Dimensional tolerances.*

Inside Dimension		Outside dimension	
Plus	Minus	Plus	Minus
Inch 1/32	*Inch* 0	*Inch* 0	*Inch* 1/32

3.10 Workmanship. — The workmanship shall be first-class. The finished gaskets shall have no breaks or creases.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection procedures. — For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.2 Qualification tests at a Government laboratory.—Qualification tests shall be conducted at a Government laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4.5.1 to 4.5.3, inclusive.

4.3 Sampling for lot acceptance tests. — Two boiler handhole gaskets from each lot of 5,000 or less, and two manhole gaskets from each lot of 1,000 or less, shall be taken for tests. These samples shall be forwarded to the Naval Engineering Experiment Station, Annapolis, Md., for chemical analysis, and comparison with the sample submitted for qualification. When an inspection lot contains less than 200 gaskets, samples of the material used in their manufacture, instead of a finished gasket, shall be taken.

4.4 Sampling for lot acceptance inspection.

4.4.1 *Sampling for intermittent procurement in isolated lots.*—A random sample of gaskets shall be selected from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on table IV in accordance with Standard MIL–STD–105.

TABLE IV.—*Sampling for visual and dimensional inspection AQL (approx.) = 1.5 percent defective.*

Number of gaskets in inspection lot	Number of gaskets in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 to 800	50	2	3
801 to 1300	75	3	4
1301 and over	110	4	5

4.4.1.1 *Sampling for continuous production.*—Where there is continuous production for the Government and inspection lots include at least 200 items, sampling for Government inspection shall be conducted in accordance with Standard MIL–STD–105 at inspection level I and AQL = 1.5 percent defective; the type of sampling to be determined by the Government inspector.

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4.4.2 *Visual and dimensional inspection.*—Each of the sample gaskets selected in accordance with table IV shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification. Any gasket in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective gaskets in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.5 Test procedures.

4.5.1 *Chemical analyses.*—Chemical analyses shall be made, using accepted laboratory methods.

4.5.2 *Class a, boiler handhole.*—Tests shall be conducted in the joints of handhole gasket test apparatus under 600 pounds pressure, 750°F. super-heated steam conditions. The apparatus shall consist of three handhole headers connected together in series. Each header shall contain 12 No. 24 oval gaskets and handhole plates, 5-7/16 by 4½ inches by ¼-inch thick. Super-heated steam at 600 pounds pressure and 750°F. shall be admitted to the top of the first drum, passed through this and the other drums, and discharged into a low-pressure steam line. At the end of each working day the valves in the steam lines shall be closed, the headers filled with boiler feed water at 140°F. and allowed to cool over night and week-ends. The test shall be continued for a period of 1,000 hours of super-heated steam conditions. The joints then shall be broken, the gaskets examined and, if in suitable condition, reinstalled for further test and reinstallation at 50-hour intervals.

4.5.3 *Class b, boiler manhole.*—Tests shall be conducted in the joint of a gasket test apparatus under conditions simulating those encountered in service. The apparatus shall consist of two high-pressure steam drums having elliptical manholes, 12 by 16 inches, and manhole plates 15-15/16 by 11-15/16 inches by 29/32-inch flange. Saturated steam at boiler pressure, 600 pounds gage, shall be admitted to the top of each drum and the condensation occurring within the drum drawn off through overflow pipe at the middle of each drum. This shall subject the upper half of each gasket to saturated steam and the lower half to water at pressures and temperatures corresponding to steam. The temperature and pressure conditions shall be maintained in the drums 8 hours daily. The drums shall be drained and the joints containing the gaskets shall be broken at frequent intervals, approximately twice a week for 5 weeks. New and unused gaskets then shall be installed for a 1,000-hour endurance run at 600 pounds gage pressure. At the end of the endurance test the joints containing the gaskets shall be broken, the gaskets examined and, if in suitable condition, reinstalled for further test and reinstallation at 50-hour intervals.

4.5.4 *Inspection tests.*—Such tests as may be necessary to determine that gaskets conform to the approval sample shall be made by the Naval Engineering Experiment Station.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Gaskets shall be securely tied or packaged 25 to a bundle.

5.2 Packing.

5.2.1 *For domestic shipment.* — Gaskets, packaged as specified in 5.1, shall be packed in nailed wood or wirebound boxes conforming to Specification NN–B–621 or NN–B–631, respectively. Not more than one size or type of gasket shall be packed in a single container. The gross weight shall not exceed 200 pounds.

5.2.2 *For overseas shipment.* — Gaskets, packaged as specified in 5.1, shall be packed in nailed wood or wirebound boxes conforming to Specification JAN–P–106 or JAN–B–107, respectively. The gross weight shall not

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exceed 150 pounds. Case liners will not be required. Shipping containers shall be strapped in accordance with the applicable container specification using flat steel strapping conforming to type I, class A or B, ¾ by 0.020 inch, of Specification QQ–S–781.

5.2.3 Procurement agencies may specify the number of bundles to be packed in any single shipping container. Not more than one size or class of gasket shall be packed in a shipping container.

5.3 Marking.—In addition to any special marking required in the contract or order, marking shall be in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Class of gasket required (see 1.2).

(c) Pressure (p. s. i.) with which gaskets are to be used (see 3.6).

(d) Size of gasket required (see 3.8.1 and 3.9.1).

6.2 In the procurement of products requiring qualification the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Military Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Bureau of Ships, Navy Department, Washington 25, D. C., and arrange to have the products that they propose to offer to the Army, the Navy, or the Air Force, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Navy Department, Washington 25, D. C.

6.3 Superseding documents. — This specification supersedes MIL–G–15342 (SHIPS). The latter superseded Navy Specification 33G11.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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